

30 D

$$A_{o,Y} = 2A_{o,X}$$

$$T_{1/2,Y} = 2T_{1/2,X}$$

$$A_X = A_{o,X} e^{-\frac{\ln 2}{T_{1/2,X}} t}$$

$$A_Y = A_{o,Y} e^{-\frac{\ln 2}{T_{1/2,Y}} t}$$

$$\frac{A_X}{A_Y} = \frac{A_{o,X} e^{-\frac{\ln 2}{T_{1/2,X}} t}}{A_{o,Y} e^{-\frac{\ln 2}{T_{1/2,Y}} t}} = \frac{A_{o,X} e^{-\frac{\ln 2}{T_{1/2,X}} t}}{2A_{o,X} e^{-\frac{\ln 2}{2T_{1/2,X}} t}} = \frac{1}{2} e^{-\frac{\ln 2}{2T_{1/2,X}} (6T_{1/2,X})} = \frac{1}{16}$$